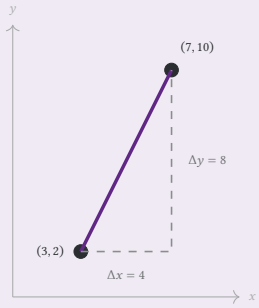
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 3

RULE · Slope (qiyalik)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Through $(3, 2)$ and $(7, 10)$:
$m = \frac{10 - 2}{7 - 3} = \frac{8}{4} = 2$

RULE · Point-slope form

$$y - y_1 = m(x - x_1)$$

Slope 4, point $(1, 3)$: $y - 3 = 4(x - 1) \implies y = 4x - 1$

Through $(2, 1)$ and $(4, 7)$: $m = \frac{7 - 1}{4 - 2} = 3 \implies y = 3x - 5$

RULES SHEET – 2/2

RULE · Parallel & perpendicular lines

$m_1 = m_2$ (parallel) $\qquad$ $m_1 \times m_2 = -1$ (perpendicular)

Perpendicular to $y = 2x - 5$, through $(4, 1)$: slope $= -\frac{1}{2} \implies y = -\frac{1}{2}x + 3$

RULE · Real-world meaning of m and b

Slope = rate of change per unit. y-intercept = starting value (at $x = 0$).

$C(x) = 2000x + 5000$ (taxi fare, som per km): $C(10) = 2000(10) + 5000 = 25000$.